Mathematical Analysis II
Week 7 Homework (April 16)

袁子强　　2025533009

2026 年 4 月 17 日

Section 9.6

8. 设 ϕ, ψ 是数量场, $\boldsymbol{a}, \boldsymbol{b}$ 为向量场, 证明

$$\begin{aligned}
\nabla(\phi+\psi) &= \nabla\phi + \nabla\psi, \\
\nabla\cdot(\boldsymbol{a}+\boldsymbol{b}) &= \nabla\cdot\boldsymbol{a} + \nabla\cdot\boldsymbol{b}, \\
\nabla\times(\boldsymbol{a}+\boldsymbol{b}) &= \nabla\times\boldsymbol{a} + \nabla\times\boldsymbol{b}, \\
\nabla(\phi\psi) &= \phi\nabla\psi + \psi\nabla\phi, \\
\nabla\cdot(\phi\boldsymbol{a}) &= \phi\nabla\cdot\boldsymbol{a} + \boldsymbol{a}\cdot\nabla\phi, \\
\nabla\cdot(\boldsymbol{a}\times\boldsymbol{b}) &= \boldsymbol{b}\cdot\nabla\times\boldsymbol{a} - \boldsymbol{a}\cdot\nabla\times\boldsymbol{b}, \\
\nabla\times(\phi\boldsymbol{a}) &= \nabla\phi\times\boldsymbol{a} + \phi\nabla\times\boldsymbol{a}.
\end{aligned}$$

证. 1. $\nabla(\phi+\psi) = (\partial_x(\phi+\psi), \partial_y(\phi+\psi), \partial_z(\phi+\psi)) = (\partial_x, \partial_y, \partial_z)(\phi+\psi) = \nabla\phi + \nabla\psi$

2. $\nabla\cdot(a+b) = \partial_x(a_1+b_1) + \partial_y(a_2+b_2) + \partial_z(a_3+b_3) = \nabla\cdot a + \nabla\cdot b$

3. $\nabla\times(a+b) = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \partial_x & \partial_y & \partial_z \\ a_1+b_1 & a_2+b_2 & a_3+b_3 \end{vmatrix} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \partial_x & \partial_y & \partial_z \\ a_1 & a_2 & a_3 \end{vmatrix} + \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \partial_x & \partial_y & \partial_z \\ b_1 & b_2 & b_3 \end{vmatrix} = \nabla\times a + \nabla\times b$

4. $\nabla(\phi\psi) = \partial_x(\phi\psi) + \partial_y(\phi\psi) + \partial_z(\phi\psi)$

   其中 $\partial_x(\phi\psi) = \phi\partial_x\psi + \psi\partial_x\phi$, ∂_y, ∂_z 同理

   所以 $\nabla(\phi\psi) = \phi(\partial_x+\partial_y+\partial_z)\psi + \psi(\partial_x+\partial_y+\partial_z)\phi = \phi\nabla\psi + \psi\nabla\phi$

5. $\nabla\cdot(\phi a) = \partial_x(\phi a) + \partial_y(\phi a) + \partial_z(\phi a)$

   其中 $\partial_x(\phi a) = \phi\partial_x a + a\partial_x\phi$, ∂_y, ∂_z 同理

   所以 $\nabla\cdot(\phi a) = \phi(\partial_x+\partial_y+\partial_z)a + a(\partial_x+\partial_y+\partial_z)\phi = \phi\nabla a + a\nabla\phi$

6. $a\times b=(a_2b_3-a_3b_2,a_3b_1-a_1b_3,a_1b_2-a_2b_1)$

所以

$$\begin{aligned}
\mathrm{LHS}&=\nabla(a\times b)\\
&=\partial_x(a_2b_3-a_3b_2)+\partial_y(a_3b_1-a_1b_3)+\partial_z(a_1b_2-a_2b_1)\\
&=b_3\partial_xa_2+a_2\partial_xb_3-a_3\partial_xb_2-b_2\partial_xa_3\\
&\quad+b_1\partial_ya_3+a_3\partial_yb_1-a_1\partial_yb_3-b_3\partial_ya_1\\
&\quad+b_2\partial_za_1+a_1\partial_zb_2-a_2\partial_zb_1-b_1\partial_za_2\\
&=b_1(\partial_ya_3-\partial_za_2)+b_2(\partial_za_1-\partial_xa_3)+b_3(\partial_xa_2-\partial_ya_1)\\
&\quad-a_1(\partial_yb_3-\partial_zb_2)-a_2(\partial_zb_1-\partial_xb_3)-a_3(\partial_xb_2-\partial_yb_1)\\
&=b(\partial_ya_3-\partial_za_2,\partial_za_1-\partial_xa_3,\partial_xa_2-\partial_ya_1)-a(\partial_yb_3-\partial_zb_2,\partial_zb_1-\partial_xb_3,\partial_xb_2-\partial_yb_1)\\
&=b\cdot\nabla\times a-a\cdot\nabla\times b\\
&=\mathrm{RHS}
\end{aligned}$$

7.

$$\begin{aligned}
\mathrm{LHS}&=\nabla\times(\phi a)\\
&=\begin{vmatrix}\boldsymbol{i}&\boldsymbol{j}&\boldsymbol{k}\\\partial_x&\partial_y&\partial_z\\\phi a_1&\phi a_2&\phi a_3\end{vmatrix}\\
&=(\partial_y\phi a_3-\partial_z\phi a_2,\partial_z\phi a_1-\partial_x\phi a_3,\partial_x\phi a_2-\partial_y\phi a_1)\\
&=(\phi\partial_ya_3+a_3\partial_y\phi-\phi\partial_za_2-a_2\partial_z\phi,\\
&\quad\phi\partial_za_1+a_1\partial_z\phi-\phi\partial_xa_3-a_3\partial_x\phi,\\
&\quad\phi\partial_xa_2+a_2\partial_x\phi-\phi\partial_ya_1-a_1\partial_y\phi)\\
&=\phi(\partial_ya_3-\partial_za_2,\partial_za_1-\partial_xa_3,\partial_xa_2-\partial_ya_1)\\
&\quad+(\partial_y\phi\cdot a_3-\partial_z\phi\cdot a_2,\partial_z\phi\cdot a_1-\partial_x\phi\cdot a_3,\partial_x\phi\cdot a_2-\partial_y\phi\cdot a_1)\\
&=\phi\nabla\times a+\nabla\phi\times a\\
&=\mathrm{RHS}
\end{aligned}$$

Q.E.D.

9. 证明

$$\operatorname{rot}\operatorname{grad}\phi=\nabla\times\nabla\phi=\mathbf{0},$$
$$\operatorname{div}\operatorname{rot}\boldsymbol{a}=\nabla\cdot(\nabla\times\boldsymbol{a})=0.$$

证. 1. $\nabla\times\nabla\phi=\begin{vmatrix}\boldsymbol{i}&\boldsymbol{j}&\boldsymbol{k}\\\partial_x&\partial_y&\partial_z\\\partial_x\phi&\partial_y\phi&\partial_z\phi\end{vmatrix}=((\partial_{yz}-\partial_{zy})\phi,(\partial_{zx}-\partial_{xz})\phi,(\partial_{xy}-\partial_{yx})\phi)=(0,0,0)=0$

2.

$$\begin{aligned}\nabla\cdot(\nabla\times a) &= (\partial_x,\partial_y,\partial_z)\cdot(\partial_y a_3-\partial_z a_2,\partial_z a_1-\partial_x a_3,\partial_x a_2-\partial_y a_1)\\ &= (\partial_{yz}-\partial_{zy})a_1+(\partial_{zx}-\partial_{xz})a_2+(\partial_{xy}-\partial_{yx})a_3\\ &= 0+0+0=0\end{aligned}$$

Q.E.D.

10. 证明

$$\nabla\times(\nabla\times\boldsymbol{v})=\nabla(\nabla\cdot\boldsymbol{v})-\nabla^2\boldsymbol{v}.$$

证.

$$\begin{aligned}\nabla\times(\nabla\times v) &= \begin{vmatrix}\boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k}\\ \partial_x & \partial_y & \partial_z\\ \partial_y v_3-\partial_z v_2 & \partial_z v_1-\partial_x v_3 & \partial_x v_2-\partial_y v_1\end{vmatrix}\\ &= (\partial_{yx}v_2-\partial_{yy}v_1-\partial_{zz}v_1+\partial_{zx}v_3,\\ &\quad \partial_{zy}v_3-\partial_{zz}v_2-\partial_{xx}v_2+\partial_{xy}v_1,\\ &\quad \partial_{xz}v_1-\partial_{xx}v_3-\partial_{yy}v_3+\partial_{yz}v_2)\\ \nabla(\nabla\cdot v) &= \nabla(\partial_x v_1+\partial_y v_2+\partial_z v_3)\\ &= (\partial_{xx}v_1+\partial_{xy}v_2+\partial_{xz}v_3,\\ &\quad \partial_{yx}v_1+\partial_{yy}v_2+\partial_{yz}v_3,\\ &\quad \partial_{zx}v_1+\partial_{zy}v_2+\partial_{zz}v_3)\\ \nabla^2 v &= (\partial_{xx}+\partial_{yy}+\partial_{zz})v\\ &= ((\partial_{xx}+\partial_{yy}+\partial_{zz})v_1,\\ &\quad (\partial_{xx}+\partial_{yy}+\partial_{zz})v_2,\\ &\quad (\partial_{xx}+\partial_{yy}+\partial_{zz})v_3)\end{aligned}$$

不难发现三个分量均满足

$$\begin{cases}(\partial_{xx}v_1+\partial_{xy}v_2+\partial_{xz}v_3)-((\partial_{xx}+\partial_{yy}+\partial_{zz})v_1)=\partial_{yx}v_2-\partial_{yy}v_1-\partial_{zz}v_1+\partial_{zx}v_3\\ (\partial_{yx}v_1+\partial_{yy}v_2+\partial_{yz}v_3)-((\partial_{xx}+\partial_{yy}+\partial_{zz})v_2)=\partial_{zy}v_3-\partial_{zz}v_2-\partial_{xx}v_2+\partial_{xy}v_1\\ (\partial_{zx}v_1+\partial_{zy}v_2+\partial_{zz}v_3)-((\partial_{xx}+\partial_{yy}+\partial_{zz})v_3)=\partial_{xz}v_1-\partial_{xx}v_3-\partial_{yy}v_3+\partial_{yz}v_2\end{cases}$$

所以 $\nabla\times(\nabla\times v)=\nabla(\nabla\cdot v)-\nabla^2 v$
Q.E.D.

第 9 章综合习题

10. 设 $f(x,y)$ 在 (x_0,y_0) 的某个邻域 U 上有定义, $\frac{\partial f}{\partial x}$ 和 $\frac{\partial f}{\partial y}$ 在 U 上存在. 求证: 如果 $\frac{\partial f}{\partial x}$ 和 $\frac{\partial f}{\partial y}$ 中有一个在 (x_0,y_0) 处连续, 那么 $f(x,y)$ 在 (x_0,y_0) 可微.

证. 不妨设 $\frac{\partial f}{\partial x}$ 在 (x_0, y_0) 处连续。

记 $\Delta f = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$，其中 $(x_0 + \Delta x, y_0 + \Delta y) \in U$。

因此

$$\Delta f = (f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y)) + (f(x_0, y_0 + \Delta y) - f(x_0, y_0)).$$

将 $f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y)$ 视为关于 x 的一元函数，由拉格朗日中值定理，存在 $\theta \in (0, 1)$ 使得

$$f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y) = \frac{\partial f}{\partial x}(x_0 + \theta \Delta x, y_0 + \Delta y)\Delta x.$$

由于 $\frac{\partial f}{\partial x}$ 在 (x_0, y_0) 处连续，因此当 $\Delta x, \Delta y \to 0$ 时，

$$\frac{\partial f}{\partial x}(x_0 + \theta \Delta x, y_0 + \Delta y) \to \frac{\partial f}{\partial x}(x_0, y_0).$$

记 $\alpha = \frac{\partial f}{\partial x}(x_0 + \theta \Delta x, y_0 + \Delta y) - \frac{\partial f}{\partial x}(x_0, y_0)$，则当 $\rho \to 0$ 时，$\alpha \to 0$。因此

$$f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0 + \Delta y) = \left(\frac{\partial f}{\partial x}(x_0, y_0) + \alpha\right)\Delta x.$$

又由于

$$f(x_0, y_0 + \Delta y) - f(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0)\Delta y + \beta \Delta y,$$

其中当 $\Delta y \to 0$ 时，$\beta \to 0$。

因此

$$\Delta f = \frac{\partial f}{\partial x}(x_0, y_0)\Delta x + \frac{\partial f}{\partial y}(x_0, y_0)\Delta y + \alpha \Delta x + \beta \Delta y,$$

其中 $|\alpha \Delta x + \beta \Delta y| \leq (|\alpha| + |\beta|)\rho \to 0 \ (\rho \to 0)$，故 $\alpha \Delta x + \beta \Delta y = o(\rho)$。

因此 $f(x, y)$ 在 (x_0, y_0) 处可微。

Q.E.D.

11. 设 $u(x, y)$ 在 $\mathbb{R}^2$ 上取正值且有二阶连续偏导数. 证明 u 满足方程

$$u\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial u}{\partial x}\frac{\partial u}{\partial y}$$

的充分必要条件是存在一元函数 f 和 g 使得 $u(x, y) = f(x)g(y)$.

证.

- **充分性：** 若存在一元函数 f 和 g 使得 $u(x, y) = f(x)g(y)$，求偏导可得

$$\begin{cases} \dfrac{\partial u}{\partial x} = f'(x)g(y), \\ \dfrac{\partial u}{\partial y} = f(x)g'(y), \\ \dfrac{\partial^2 u}{\partial x \partial y} = f'(x)g'(y). \end{cases}$$

因此

$$\text{LHS} = f(x)g(y)f'(x)g'(y),$$

$$\text{RHS} = (f'(x)g(y))(f(x)g'(y)) = f(x)g(y)f'(x)g'(y) = \text{LHS}.$$

- **必要性：** 由于 $u\frac{\partial^2 u}{\partial x\partial y}=\frac{\partial u}{\partial x}\frac{\partial u}{\partial y}$，可将其改写为

$$\frac{u\frac{\partial^2 u}{\partial x\partial y}-\frac{\partial u}{\partial x}\frac{\partial u}{\partial y}}{u^2}=0.$$

注意到

$$\frac{u\frac{\partial^2 u}{\partial x\partial y}-\frac{\partial u}{\partial x}\frac{\partial u}{\partial y}}{u^2}=\frac{\partial}{\partial y}\left(\frac{\frac{\partial u}{\partial x}}{u}\right)=\frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}\ln u\right),$$

因此

$$\frac{\partial}{\partial y}\left(\frac{\partial}{\partial x}\ln u\right)=0.$$

从而 $\frac{\partial}{\partial x}\ln u=A(x)$，积分可得

$$\ln u=\int A(x)\,\mathrm{d}x+B(y).$$

记 $\alpha(x)=\int A(x)\,\mathrm{d}x$ 为关于 x 的一元函数，则

$$u=\mathrm{e}^{\alpha(x)+B(y)}=\mathrm{e}^{\alpha(x)}\mathrm{e}^{B(y)}.$$

令 $f(x)=\mathrm{e}^{\alpha(x)}$，$g(y)=\mathrm{e}^{B(y)}$，则存在一元函数 f 和 g 使得 $u(x,y)=f(x)g(y)$。

Q.E.D.

12. 设 $\boldsymbol{r}(t)=x(t)\boldsymbol{i}+y(t)\boldsymbol{j}$, $t\in[a,b]$ 有连续的导数, 证明: 存在 $\theta\in(a,b)$, 使得

$$|\boldsymbol{r}(b)-\boldsymbol{r}(a)|\le|\boldsymbol{r}'(\theta)|(b-a).$$

提示: 令 $\varphi(t)=(x(b)-x(a))x(t)+(y(b)-y(a))y(t)$, 对函数 $\varphi(t)$ 使用微分中值定理, 并利用 Cauchy-Schwarz 不等式: $(a_1b_1+a_2b_2)^2\le(a_1^2+a_2^2)(b_1^2+b_2^2)$.

证. 令 $\varphi(t)=(x(b)-x(a))x(t)+(y(b)-y(a))y(t)$。由于 $x(t),y(t)$ 在 $[a,b]$ 上导数连续，因此 $\varphi(t)$ 在 $[a,b]$ 上连续，在 (a,b) 上可导。

由拉格朗日中值定理，存在 $\theta\in(a,b)$ 使得

$$\varphi(b)-\varphi(a)=\varphi'(\theta)(b-a).$$

其中

$$\begin{aligned}\varphi(b)-\varphi(a)&=(x(b)-x(a))x(b)+(y(b)-y(a))y(b)-(x(b)-x(a))x(a)-(y(b)-y(a))y(a)\\&=-x(b)x(a)+x^2(a)-y(b)y(a)+y^2(a)+x^2(b)-x(a)x(b)+y^2(b)-y(a)y(b)\\&=(x(b)-x(a))^2+(y(b)-y(a))^2\\&=|\boldsymbol{r}(b)-\boldsymbol{r}(a)|^2.\end{aligned}$$

又由于

$$\begin{aligned}(\varphi'(\theta))^2 &= [(x(b)-x(a))x'(\theta)+(y(b)-y(a))y'(\theta)]^2\\ &\le [(x(b)-x(a))^2+(y(b)-y(a))^2][(x'(\theta))^2+(y'(\theta))^2]\\ &= |\boldsymbol{r}(b)-\boldsymbol{r}(a)|^2|\boldsymbol{r}'(\theta)|^2,\end{aligned}$$

因此

$$\varphi'(\theta)\le |\boldsymbol{r}(b)-\boldsymbol{r}(a)||\boldsymbol{r}'(\theta)|.$$

从而

$$|\boldsymbol{r}(b)-\boldsymbol{r}(a)|^2=\varphi'(\theta)(b-a)\le |\boldsymbol{r}(b)-\boldsymbol{r}(a)||\boldsymbol{r}'(\theta)|(b-a).$$

若 $|\boldsymbol{r}(b)-\boldsymbol{r}(a)|=0$，则 $|\boldsymbol{r}(b)-\boldsymbol{r}(a)|\le|\boldsymbol{r}'(\theta)|(b-a)$ 显然成立。

若 $|\boldsymbol{r}(b)-\boldsymbol{r}(a)|\ne 0$，则 $|\boldsymbol{r}(b)-\boldsymbol{r}(a)|\le|\boldsymbol{r}'(\theta)|(b-a)$。

Q.E.D.

14. 求函数 $f(x,y)=x^2+xy^2-x$ 在区域 $D=\{(x,y)\mid x^2+y^2\le 2\}$ 上的最大值和最小值.

解. 首先计算偏导数：

$$\begin{cases} f_x=2x+y^2-1=0,\\ f_y=2xy=0.\end{cases}$$

解得驻点

$$\begin{cases} x=0,\\ y=\pm1,\end{cases}\qquad \text{或}\qquad \begin{cases} x=\dfrac{1}{2},\\ y=0,\end{cases}$$

均满足 $x^2+y^2\le 2$，位于定义域内。

对应的函数值为

$$\begin{cases} f(0,\pm1)=0,\\ f\left(\dfrac{1}{2},0\right)=-\dfrac{1}{4}.\end{cases}$$

接下来考虑边界条件，约束条件为 $x^2+y^2-2=0$。

构造拉格朗日函数

$$L(x,y,\lambda)=x^2+xy^2-x-\lambda(x^2+y^2-2).$$

计算偏导数：

$$\begin{cases} L_x=2x+y^2-1-2\lambda x=0,\\ L_y=2xy-2\lambda y=0,\\ L_\lambda=-x^2-y^2+2=0.\end{cases}$$

解得边界上的候选点

$$\begin{cases} x = \pm\sqrt{2}, \\ y = 0, \end{cases} \quad \text{或} \quad \begin{cases} x = 1, \\ y^2 = 1, \end{cases} \quad \text{或} \quad \begin{cases} x = -\dfrac{1}{3}, \\ y^2 = \dfrac{17}{9}. \end{cases}$$

对应的函数值为

$$\begin{cases} f(\sqrt{2}, 0) = 2 - \sqrt{2}, \\ f(-\sqrt{2}, 0) = 2 + \sqrt{2}, \\ f(1, \pm 1) = 1, \\ f\left(-\dfrac{1}{3}, \pm\dfrac{\sqrt{17}}{3}\right) = -\dfrac{5}{27}. \end{cases}$$

比较所有候选点的函数值，显然 $2+\sqrt{2}$ 最大，$-\dfrac{1}{4}$ 最小。

因此最大值为 $f(-\sqrt{2}, 0) = 2 + \sqrt{2}$，最小值为 $f\left(\dfrac{1}{2}, 0\right) = -\dfrac{1}{4}$。

15. 设 $x_1, x_2, \ldots, x_n$ 是正数, 且 $x_1 + x_2 + \cdots + x_n = n$. 用 Lagrange 乘数法证明

$$x_1 x_2 \cdots x_n \left(\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}\right) \le n,$$

等号当且仅当 $x_1 = x_2 = \cdots = x_n = 1$ 时成立.

证. 记 $F(x_1, \ldots, x_n) = \sum_{i=1}^n \prod_{j\neq i} x_j$ 即为左端表达式，约束条件为 $\sum_{i=1}^n x_i = n$。

构造拉格朗日函数

$$L = \sum_{i=1}^n \prod_{j\neq i} x_j + \lambda \left(\sum_{i=1}^n x_i - n\right).$$

计算偏导数：

$$\begin{cases} \dfrac{\partial L}{\partial x_k} = \left(\prod_{j\neq k} x_j\right)\left(\sum_{i\neq k} \dfrac{1}{x_i}\right) + \lambda = 0, \\ L_\lambda = \sum_{i=1}^n x_i - n = 0. \end{cases}$$

即

$$\left(\prod x_i\right)\left(\sum_{i\neq k} \frac{1}{x_i}\right) = -\lambda x_k.$$

因此对于任意 $1 \le k, l \le n$，有

$$\frac{\sum_{i\neq k} \frac{1}{x_i}}{\sum_{i\neq l} \frac{1}{x_i}} = \frac{x_k}{x_l}.$$

从而

$$x_l \left(\left(\sum \frac{1}{x_i}\right) - \frac{1}{x_k}\right) = x_k \left(\left(\sum \frac{1}{x_i}\right) - \frac{1}{x_l}\right),$$

即

$$\left(\sum \frac{1}{x_i}\right)(x_l - x_k) = \frac{x_l^2 - x_k^2}{x_k x_l} = (x_l - x_k)\left(\frac{1}{x_k} + \frac{1}{x_l}\right).$$

若 $x_l \neq x_k$，记 $S = \sum \frac{1}{x_i}$，则 $\sum_{i\neq k,l} \frac{1}{x_i} = 0$，这与 $x_1, x_2, \ldots, x_n$ 均为正数矛盾。因此对任意 k, l 均有 $x_l = x_k = \dfrac{n}{n} = 1$。

当任意 $x_i \to 0^+$ 时，$F \to 0^+$，因此该点为最大值点。

从而

$$x_1 x_2 \cdots x_n \left(\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}\right) \leq n.$$

当 $n = 2$ 时，等号恒成立；当 $n > 2$ 时，等号成立当且仅当 $x_1 = x_2 = \cdots = x_n = 1$。

Q.E.D.

16. 设 $a_i \geq 0$ $(i = 1, 2, \ldots, n)$, $p > 1$. 证明:

$$\frac{a_1 + \cdots + a_n}{n} \leq \left(\frac{a_1^p + \cdots + a_n^p}{n}\right)^{1/p},$$

并讨论等号成立的条件.

证. 即证

$$\left(\frac{a_1 + \cdots + a_n}{n}\right)^p \leq \frac{a_1^p + \cdots + a_n^p}{n}.$$

令 $f(x) = x^p$（$x \geq 0$），则 $f'(x) = px^{p-1}$，$f''(x) = p(p-1)x^{p-2}$。由于 $p > 0$，$p - 1 > 0$，$x^{p-2} \geq 0$，因此 $f''(x) \geq 0$，故 $f(x)$ 为凸函数。

由于 $\frac{1}{n}$ 非负，由 Jensen 不等式可得

$$f\left(\frac{a_1 + \cdots + a_n}{n}\right) \leq \frac{f(a_1) + \cdots + f(a_n)}{n},$$

等号成立当且仅当 $a_1 = \cdots = a_n$。

即

$$\frac{a_1 + \cdots + a_n}{n} \leq \left(\frac{a_1^p + \cdots + a_n^p}{n}\right)^{1/p},$$

等号成立当且仅当 $a_1 = \cdots = a_n$。

Q.E.D.